Section 2.2 Areas & area ratios

面积与面积比

Method 1 / 方法 1 [GE02-M01]

Area often eliminates an unknown height or point. Use common heights for ratios, squared scales for similarity, determinants for coordinates and sectors for circular pieces. Preserve signed areas during algebra, then assemble the ordinary region without overlap.

面积常能消去未知高度或点。等高用比例，相似用平方比，坐标用行列式，圆块用扇形。代数中保留有向面积，最后再无重叠地拼成普通区域。

Common-height ratios and corner subtraction eliminate shape

等高比例与扣角消去形状

Procedure

1. Triangles with a common altitude have area ratio equal to their base ratio.

2. With a common included angle, area ratio is the product of the two adjacent-side ratios.

3. For an inner triangle, subtract three corner triangles using consistent cyclic side fractions.

4. Only cancel heights after verifying the bases lie on the intended common line.

通用步骤

1. 共高三角形面积比等于底边比。

2. 共夹角三角形面积比等于两邻边比例的乘积。

3. 内部三角形可通过一致的循环边分数扣除三个角三角形。

4. 先确认各底在所需同一直线上，再消去高度。

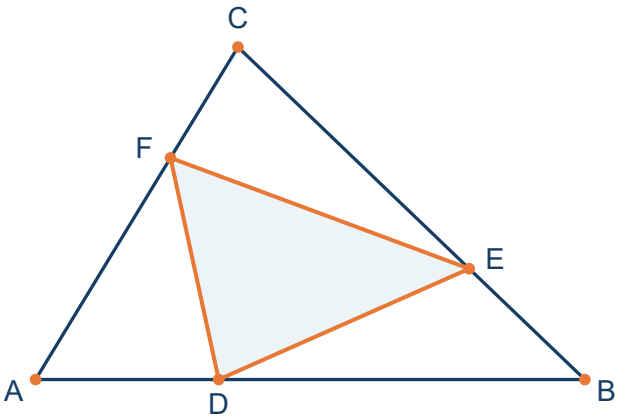
> Example. Points D, E, F lie respectively one third of the way along AB, BC, CA. Find $[DEF]/[ABC]$.

Section 2.2 Areas & area ratios

面积与面积比

Method 1 / 方法 1 [GE02-M01] (continued / 续)

> 例题：D, E, F 分别位于沿 AB, BC, CA 前进三分之一处，求 $[DEF]/[ABC]$。



AD/AB = BE/BC = CF/CA = ⅓

Worked solution

1. At A, the corner triangle has side fractions $1/3$ on AB and $2/3$ on AC, so its area fraction is $2/9$.

2. The other two corner fractions are also $2/9$. Their interiors are disjoint.

3. The inner fraction is $1 - 3(2/9) = 1/3$.

例题解析

1. A 角三角形在 AB, AC 上的边比例为 $1/3, 2/3$，面积占比为 $2/9$。

2. 另两角面积占比同为 $2/9$，且内部不交。

3. 内部比例为 $1 - 3(2/9) = 1/3$。

For cyclic fractions p, q, r, the inner area fraction is $1 - p - q - r + pq + qr + rp$.

Section 2.2 Areas & area ratios

面积与面积比

Method 1 / 方法 1 [GE02-M01] (continued / 续)

循环边分数为 p, q, r 时，内部面积占比为 $1 - p - q - r + pq + qr + rp$。

Practice / 配套练习: AIME-2001-I-Q09 (WB p. 247), AIME-2002-I-Q10 (WB p. 314), AIME-2002-I-Q13 (WB p. 293), AIME-2004-II-Q13 (WB p. 232), AIME-2007-I-Q15 (WB p. 234), AIME-2010-I-Q15 (WB p. 320), AIME-2013-II-Q05 (WB p. 260), AIME-2015-II-Q15 (WB p. 345), AIME-2017-II-Q10 (WB p. 377), AIME-2018-II-Q12 (WB p. 280)

Section 2.2 Areas & area ratios

面积与面积比

Method 2 / 方法 2 [GE02-M02]

Similarity squares linear scales before areas are subtracted

相似先平方线性比再扣面积

Procedure

1. Prove the triangles are similar and determine their positive linear scale.

2. Square that scale for area; keep the whole and complementary pieces separate.

3. For a sequence of parallel bands, compare cumulative area from the same apex.

4. For an infinite construction, identify the area ratio before summing the geometric series.

通用步骤

1. 先证明相似并求正线性比例。

2. 面积取该比例平方，分别记录整体与补块。

3. 平行条带应比较从同一顶点量起的累积面积。

4. 无穷构造先确定面积公比，再求等比级数。

> Example. A triangle has area 72. Two parallel chords lie at one half and one third of its altitude measured from the apex. Find the area of the band between them.

> 例题：一个三角形面积为72。两条平行截线从顶点量起的高度，分别是原高的一半与三分之一。求两截线之间条带的面积。

Worked solution

1. The two apex triangles have linear scales $1/2$ and $1/3$, since all three bases are parallel.

Section 2.2 Areas & area ratios

面积与面积比

Method 2 / 方法 2 [GE02-M02] (continued / 续)

2. Their areas are $72(1/2)^2 = 18$ and $72(1/3)^2 = 8$.

3. The smaller apex triangle is contained in the larger, so the band area is $18 - 8 = 10$, or $5/36$ of the original area.

例题解析

1. 三底平行，所以两个顶端小三角形的线性比例分别为 $1/2, 1/3$。

2. 两面积分别为 $72(1/2)^2 = 18$ 与 $72(1/3)^2 = 8$。

3. 较小顶端三角形包含在较大者内，故条带面积为 $18 - 8 = 10$，即原面积的 $5/36$。

Half the height does not generally mean half the area.

一半高度通常不对应一半面积。

For a trapezoid of parallel bases $0 < a < b$, extend its legs to an apex. A parallel chord x bisects the trapezoid area exactly when $x^2 - a^2 = b^2 - x^2$, hence $x = \sqrt{(a^2 + b^2)/2}$. The positive root lies between the two bases.

两底满足 $0 < a < b$ 的梯形，延长两腰至公共顶点。平行截段 x 平分面积，等价于 $x^2 - a^2 = b^2 - x^2$，所以 $x = \sqrt{(a^2 + b^2)/2}$，正根介于两底之间。

Practice / 配套练习: AIME-2000-II-Q06 (WB p. 246), AIME-2003-II-Q06 (WB p. 248), AIME-2004-I-Q09 (WB p. 225), AIME-2006-I-Q07 (WB p. 249), AIME-2006-II-Q12 (WB p. 233), AIME-2007-I-Q12 (WB p. 253), AIME-2008-I-Q02 (WB p. 241), AIME-2010-I-Q13 (WB p. 255), AIME-2010-II-Q09 (WB p. 372), AIME-2011-I-Q08 (WB p. 251), AIME-2013-I-Q13 (WB p. 235), AIME-2015-I-Q07 (WB p. 375), AIME-2021-II-Q02 (WB p. 244)

Section 2.2 Areas & area ratios

面积与面积比

Method 3 / 方法 3 [GE02-M03]

Determinants measure simple polygons and spatial triangles

行列式度量简单多边形与空间三角形

Procedure

1. List a simple polygon’s vertices in boundary order before using the shoelace formula.

2. Compute twice the signed area as the sum of consecutive cross products, taking absolute value only at the end.

3. For a spatial triangle, use half the norm of the two side-vector cross product.

4. For a self-intersecting boundary, split its bounded regions and add ordinary areas instead of taking one global shoelace absolute value.

通用步骤

1. 用鞋带公式前，按边界顺序列出简单多边形顶点。

2. 相邻点叉积和为两倍有向面积，最后才取绝对值。

3. 空间三角形用两边向量叉积长度的一半。

4. 自交边界应分割各有界区域并加普通面积，不能对全局鞋带结果直接取绝对值。

> Example. Find the area of the concave polygon $(0,0),(4,0),(4,1),(1,1),(1,3),(0,3)$.

> 例题：求凹多边形 $(0,0),(4,0),(4,1),(1,1),(1,3),(0,3)$ 的面积。

Worked solution

1. The consecutive cross products are $0,4,3,2,3,0$, summing to twelve.

Section 2.2 Areas & area ratios

面积与面积比

Method 3 / 方法 3 [GE02-M03] (continued / 续)

2. The area is $|12|/2 = 6$.

3. As a check, subtract a 3×2 notch from a 4×3 rectangle: $12 - 6 = 6$.

例题解析

1. 相邻叉积依次为 $0, 4, 3, 2, 3, 0$，总和十二。

2. 面积为 $|12|/2 = 6$。

3. 也可从四乘三矩形扣三乘二缺口，得到 $12 - 6 = 6$。

The signed area of a crossed polygon may cancel bounded lobes.

交叉多边形的有向面积可能把部分有界叶片相互抵消。

For lattice points on $ax + y = N$, triangle area with the origin is $N|i - j|/2$ after parameterizing points by their integer x coordinates. Integrality can therefore become a parity condition.

An affine transformation multiplies every signed area by the same determinant. Normalize a parallelogram to the unit square when only area ratios are needed.

将 $ax + y = N$ 上格点按整数横坐标编号后，与原点组成的面积为 $N|i - j|/2$，面积整性因而可能转化为奇偶条件。

仿射变换把全部有向面积乘同一个行列式。只求面积比时，可把平行四边形规范为单位正方形。

Section 2.2 Areas & area ratios

面积与面积比

Method 3 / 方法 3 [GE02-M03] (continued / 续)

Practice / 配套练习: AIME-2000-I-Q13 (WB p. 384), AIME-2003-II-Q14 (WB p. 389), AIME-2005-I-Q10 (WB p. 368), AIME-2009-I-Q11 (WB p. 254), AIME-2013-I-Q07 (WB p. 373), AIME-2013-II-Q13 (WB p. 399), AIME-2014-I-Q13 (WB p. 400), AIME-2015-I-Q04 (WB p. 357), AIME-2017-II-Q03 (WB p. 767), AIME-2017-II-Q10 (WB p. 377), AIME-2018-II-Q04 (WB p. 358), AIME-2018-II-Q09 (WB p. 379), AIME-2019-II-Q01 (WB p. 359), AIME-2025-I-Q02 (WB p. 245), AIME-2025-I-Q12 (WB p. 413), AIME-2025-II-Q01 (WB p. 364), AIME-2026-I-Q10 (WB p. 383), AIME-2026-II-Q03 (WB p. 365)

Section 2.2 Areas & area ratios

面积与面积比

Method 4 / 方法 4 [GE02-M04]

Sectors, segments and annuli assemble circular area

扇形、弓形及圆环拼成圆面积

Procedure

1. A circular segment is a sector minus its central triangle; specify the minor or complementary part.

2. Annulus area is $\pi(R^2 - r^2)$, so consecutive radius squares often telescope or pair.

3. For a symmetric disk union, assign each angular wedge to the disk reaching farthest along its rays.

4. Confirm all subtractions or unions are disjoint except at boundaries.

通用步骤

1. 圆弓形为扇形减圆心三角形，先明确劣弓形或补集。

2. 圆环面积为 $\pi(R^2 - r^2)$，连续半径平方常可配对或消项。

3. 对称圆盘并集可按射线伸得最远的圆盘分配角扇区。

4. 核实各扣块或拼块除边界外互不重叠。

> Example. A chord in a radius-two circle is distance one from its center. Find the smaller circular segment area.

> 例题：半径二的圆中，一弦距圆心为一，求较小弓形面积。

Section 2.2 Areas & area ratios

面积与面积比

Method 4 / 方法 4 [GE02-M04] (continued / 续)

Worked solution

1. The half central angle has cosine $1/2$, so the full angle is 120°.

2. The sector area is $4\pi/3$, and its isosceles central triangle has area $\frac{1}{2}(2)(2)\sin 120^\circ = \sqrt{3}$.

3. The smaller segment area is $4\pi/3 - \sqrt{3}$.

例题解析

1. 中心半角余弦为 $1/2$，所以中心角为 120°。

2. 扇形面积为 $4\pi/3$，圆心等腰三角形面积为 $\frac{1}{2}(2)(2)\sin 120^\circ = \sqrt{3}$。

3. 小弓形面积为 $4\pi/3 - \sqrt{3}$。

The sum of disk areas counts overlaps repeatedly.

直接相加各圆盘面积会重复计算重叠。

Six unit disks centered at sixth roots of unity have a union radial boundary $2\cos\theta$ on each 60° wedge. Each wedge area is $\pi/3 + \sqrt{3}/2$, obtained from a 120° unit sector and two triangles.

六个以六次单位根为圆心的单位圆盘，其并集在每个 60° 扇区内的径向边界为 $2\cos\theta$。每区面积为 $\pi/3 + \sqrt{3}/2$，可分为一个 120° 单位扇形与两个三角形。

Practice / 配套练习: AIME-2003-I-Q02 (WB p. 237), AIME-2004-I-Q04 (WB p. 238), AIME-2004-II-Q01 (WB p. 239), AIME-2005-I-Q01 (WB p. 329), AIME-2008-II-Q13 (WB p. 90), AIME-2010-I-Q13 (WB p. 255), AIME-2015-I-Q15 (WB p. 459), AIME-2026-I-Q03 (WB p. 424)

Section 2.2 Areas & area ratios

面积与面积比

Method 5 / 方法 5 [GE02-M05]

Regular polygons and hinged shapes reduce to triangle pieces

正多边形与铰接图形化为三角形块

Procedure

1. Decompose a regular polygon, rhombus or trapezoid into triangles with known angles or common heights.

2. Use $ab\sin\theta/2$ for each piece and count cell rings before multiplying a cell area.

3. For a hinged deformation recover the perpendicular height from the rigid bar lengths.

4. If the shape varies, state its full angle interval so strict endpoints and attainable areas are distinguished.

通用步骤

1. 将正多边形、菱形或梯形分成已知角或共高三角形。

2. 每块用 $ab\sin\theta/2$，蜂窝环先计格数再乘单位格面积。

3. 铰接形变先由固定杆长求垂直高度。

4. 活动构型应列完整角区间，区分严格端点与可达面积。

> Example. A rhombus has side two and acute angle $0<\theta<90^\circ$. Which positive integer areas are attainable?

> 例题：菱形边长二，锐角满足 $0<\theta<90^\circ$，哪些正整数面积可达?

Section 2.2 Areas & area ratios

面积与面积比

Method 5 / 方法 5 [GE02-M05] (continued / 续)

Worked solution

1. Its area is $4\sin\theta$. Since sine increases continuously from zero to one on this open interval, area ranges over $(0, 4)$.

2. The positive integers in that interval are one, two and three.

3. Each is attained by $\theta = \arcsin(K/4)$; area four requires the excluded right angle.

例题解析

1. 面积为 $4\sin\theta$。正弦在该开区间连续严格增，从而面积遍历 $(0, 4)$。

2. 其中正整数为一、二、三。

3. 各值均由 $\theta = \arcsin(K/4)$ 实现，面积四需要被排除的直角。

An endpoint square may be excluded even when approached arbitrarily closely.

即使能任意接近正方形，端点构型仍可能被排除。

A regular unit-side hexagon has area $3\sqrt{3}/2$. A cluster's jth ring contains $6j$ cells; sum rings before subtracting an outer paved layer.

In a symmetric five-rhombus assembly with fixed outer area C and outer meeting angle α, a straight-angle condition can make the central area $2C\cos\alpha$. Check the entire nondegenerate angle range before counting integer areas.

An equilateral triangle has area $\sqrt{3}s^2/4$. A trapezoid with parallel bases a, b and perpendicular height h has area $(a+b)h/2$; do not use a slanted leg as height.

Section 2.2 Areas & area ratios

面积与面积比

Method 5 / 方法 5 [GE02-M05] (continued / 续)

边长一的正六边形面积为 $3\sqrt{3}/2$。蜂窝簇第 j 环有 $6j$ 格，应先求环格总数，再扣最外层。

对称五菱形拼图中，若外块面积固定为 C、外块相接角为 α，平角条件可能使中央面积为 $2C\cos\alpha$；计整数面积前须检查全部非退化角范围。

等边三角形面积为 $\sqrt{3}s^2/4$。两平行底为 a, b、垂直高为 h 的梯形面积为 $(a+b)h/2$，不能以斜腰代替高度。

Practice / 配套练习: AIME-2002-II-Q04 (WB p. 236), AIME-2006-I-Q08 (WB p. 250), AIME-2006-II-Q01 (WB p. 240), AIME-2012-I-Q13 (WB p. 396), AIME-2013-I-Q03 (WB p. 14), AIME-2013-I-Q12 (WB p. 398), AIME-2014-II-Q03 (WB p. 242), AIME-2014-II-Q10 (WB p. 78), AIME-2019-I-Q03 (WB p. 243), AIME-2021-I-Q09 (WB p. 380), AIME-2023-II-Q09 (WB p. 288), AIME-2025-I-Q06 (WB p. 290), AIME-2025-II-Q06 (WB p. 339), AIME-2025-II-Q14 (WB p. 256)

Section 2.2 Areas & area ratios

面积与面积比

Method 6 / 方法 6 [GE02-M06]

Choose between sine area, Heron and inradius area

在正弦面积、海伦与内切圆面积中选择

Procedure

1. Use $K = ab\sin C/2$ when two sides and their included angle are known.

2. Use Heron for three sides, or $K = rs$ when inradius and semiperimeter are available.

3. Equate formulas to eliminate an unknown angle, radius or scale.

4. A fixed sine usually has acute and obtuse angle branches; test both against the original conditions.

通用步骤

1. 两边及夹角已知时用 $K = ab\sin C/2$。

2. 三边已知用海伦，内切圆半径与半周长已知用 $K = rs$。

3. 联立公式消去未知角、半径或尺度。

4. 正弦值固定通常有锐角与钝角两分支，应按原条件逐一检验。

> Example. Two triangle sides are three and five and the area is six. Find both possible third-side lengths.

> 例题：三角形两边为三与五，面积为六，求第三边的两种可能长度。

Worked solution

1. The included angle satisfies $6 = (15/2)\sin C$, so $\sin C = 4/5$.

Section 2.2 Areas & area ratios

面积与面积比

Method 6 / 方法 6 [GE02-M06] (continued / 续)

2. The two possible cosines are $3/5, -3/5$. Cosine law gives third-side squares $34 - 30(3/5) = 16$ and $34 + 18 = 52$.

3. The possible lengths are 4 and $2\sqrt{13}$, both satisfying the strict triangle inequalities with three and five.

例题解析

1. 夹角满足 $6 = (15/2)\sin C$，故 $\sin C = 4/5$。

2. 余弦可为 $3/5, -3/5$，余弦定理给出第三边平方为 $16, 52$。

3. 所以第三边为 4 或 $2\sqrt{13}$，两者均与三、五满足严格三角不等式。

Equal area and two fixed sides do not necessarily give congruent triangles.

同面积及两固定边，不一定确定全等三角形。

Practice / 配套练习: AIME-2005-I-Q15 (WB p. 294), AIME-2006-II-Q15 (WB p. 275), AIME-2009-I-Q12 (WB p. 319), AIME-2010-II-Q12 (WB p. 278), AIME-2020-II-Q13 (WB p. 347), AIME-2021-II-Q05 (WB p. 265), AIME-2024-I-Q14 (WB p. 462), AIME-2024-II-Q10 (WB p. 317)

Section 2.2 Areas & area ratios

面积与面积比

Method 7 / 方法 7 [GE02-M07]

Area differences are linear normal projections

面积差是法向线性投影

Procedure

1. Orient a fixed base and write its signed triangle area as half base length times signed perpendicular distance.

2. When a circular arc is fixed, its region area differs from the central sector by this linear projection term.

3. Add normal-vector identities to eliminate an unknown point.

4. For an interior point of an equilateral triangle, the three positive side distances sum to its altitude.

通用步骤

1. 给固定底边定向，将有向三角面积写为底长一半乘有向垂距。

2. 固定圆弧时，相应区域面积与圆心扇形的差就是这个线性投影项。

3. 把法向量恒等式相加以消去未知点。

4. 等边三角形内部点到三边的正距离和等于三角形高。

> Example. An equilateral triangle has altitude six. An interior point has distances one and two to two sides. Find its distance to the third side.

> 例题：等边三角形高六，内部点到其中两边距离为一、二，求到第三边的距离。

Section 2.2 Areas & area ratios

面积与面积比

Method 7 / 方法 7 [GE02-M07] (continued / 续)

Worked solution

1. Join the point to all three vertices. The three smaller triangles have a common base length equal to the equilateral side a.

2. Area addition gives $a(d_1 + d_2 + d_3)/2 = a \cdot 6/2$.

3. Hence $d_3 = 6 - 1 - 2 = 3$. Positivity agrees with the interior-point condition.

例题解析

1. 连点与三顶点，三个小三角形可分别以等边长 a 为底。

2. 面积相加给出 $a(d_1 + d_2 + d_3)/2 = a \cdot 6/2$。

3. 故 $d_3 = 3$，其为正，符合内部点条件。

Outside the triangle, use signed distances rather than three positive distances.

点在三角形外时，应使用有向距离，而不是三个正距离。

For a regular polygon in a circle, let u_j be the outward unit normal toward side j and a its side length. A fixed-arc region at P has area equal to its central sector minus $(a/2)P \cdot u_j$. Linear relations among normals give exact relations among region areas.

圆内正多边形第 j 边的外法向单位向量为 u_j、边长为 a 时，点 P 对应固定弧区域面积等于圆心扇形面积减 $(a/2)P \cdot u_j$。法向量的线性关系因而给出区域面积的精确关系。

Practice / 配套练习: AIME-2019-II-Q13 (WB p. 407), AIME-2023-I-Q12 (WB p. 411)

Section 2.2 Areas & area ratios

面积与面积比

Method 8 / 方法 8 [GE02-M08]

Diagonals encode quadrilateral area and angle

对角线编码四边形面积及夹角

Procedure

1. For convex $ABCD$, orient the full diagonals as $u = \overrightarrow{AC}$ and $v = \overrightarrow{BD}$; then $K = |u \times v|/2$.

2. The side-square difference satisfies $BC^2 + DA^2 - AB^2 - CD^2 = 2u \cdot v$.

3. Divide cross and dot magnitudes to obtain the acute diagonal-angle tangent.

4. In a cyclic quadrilateral, Ptolemy and Brahmagupta supply the diagonal product and area separately.

通用步骤

1. 对凸四边形 $ABCD$，将整条对角线定向为 $u = \overrightarrow{AC}, v = \overrightarrow{BD}$，则 $K = |u \times v|/2$。

2. 边长平方差满足 $BC^2 + DA^2 - AB^2 - CD^2 = 2u \cdot v$。

3. 以叉积、点积绝对值之比得到对角线锐夹角的正切。

4. 共圆时，托勒密与婆罗摩笈多可分别提供对角线乘积及面积。

> Example. A convex quadrilateral has diagonals six and eight meeting at 30°. Find its area and the magnitude of their dot product.

> 例题：凸四边形对角线长六、八，夹角为 30°，求面积及点积绝对值。

Section 2.2 Areas & area ratios

面积与面积比

Method 8 / 方法 8 [GE02-M08] (continued / 续)

Worked solution

1. The diagonal cross-product magnitude is $6 \cdot 8 \sin 30^\circ = 24$.

2. Area is half that, 12.

3. The dot-product magnitude is $48 \cos 30^\circ = 24\sqrt{3}$, checking the angle tangent $24/(24\sqrt{3}) = 1/\sqrt{3}$.

例题解析

1. 对角向量叉积长度为 $6 \cdot 8 \sin 30^\circ = 24$。

2. 面积为其一半，即十二。

3. 点积绝对值为 $24\sqrt{3}$，两者之比 $1/\sqrt{3}$ 核实夹角。

Use full diagonals, not just the two pieces next to their intersection.

应使用整条对角线，不是交点旁的两个分段。

Practice / 配套练习: AIME-2018-I-Q15 (WB p. 300), AIME-2018-II-Q12 (WB p. 280), AIME-2021-I-Q11 (WB p. 306), AIME-2021-II-Q12 (WB p. 410)

Section 2.2 Areas & area ratios

面积与面积比

Method 9 / 方法 9 [GE02-M09]

Projection factors convert planar strips into spatial section areas

投影系数将平面条带化为空间截面积

Procedure

1. Locate every section vertex or curved boundary before projecting.

2. Compute the projected polygon or disk-strip area exactly.

3. For section plane $z = px + qy + c$, multiply projected area by $\sqrt{1 + p^2 + q^2}$; equivalently divide by the normal-angle cosine.

通用步骤

1. 投影前先定位全部截面顶点或曲边。

2. 精确求投影多边形或圆盘条带面积。

3. 平面 $z = px + qy + c$ 的截面积等于投影面积乘 $\sqrt{1 + p^2 + q^2}$，即除以法向夹角余弦。

> Example. Find the area of the patch $z = 2x$ above $0 \le x \le 3, 0 \le y \le 2$.

> 例题：求平面 $z = 2x$ 在 $0 \le x \le 3, 0 \le y \le 2$ 上方截片的面积。

Worked solution

1. The horizontal projection is a three-by-two rectangle, area six.

2. The plane has $p = 2, q = 0$, so the enlargement factor is $\sqrt{1 + 4} = \sqrt{5}$.

3. The patch area is $6\sqrt{5}$, also checked by adjacent edge vectors $(3, 0, 6), (0, 2, 0)$.

Section 2.2 Areas & area ratios

面积与面积比

Method 9 / 方法 9 [GE02-M09] (continued / 续)

例题解析

1. 水平投影为三乘二矩形，面积六。

2. 平面有 $p = 2, q = 0$，放大系数为 $\sqrt{5}$。

3. 截片面积为 $6\sqrt{5}$，亦可由相邻边向量 $(3, 0, 6), (0, 2, 0)$ 核实。

A finite cylinder may project to a clipped strip rather than a full disk.

有限圆柱截面可能投影为裁剪条带，而非整圆盘。

Practice / 配套练习: AIME-2007-I-Q13 (WB p. 456), AIME-2015-I-Q15 (WB p. 459)

Section 2.2 Areas & area ratios

面积与面积比

Method 10 / 方法 10 [GE02-M10]

Area bounds require a matching geometric construction

面积界必须配套几何构造

Procedure

1. Express the target as a constant times a triangle area or sine of one angle.

2. Use the area bound for fixed adjacent sides, or a containment comparison of altitudes.

3. Identify the exact equality angle and prove it lies in the original feasible family.

通用步骤

1. 把目标写成某三角面积或一个角正弦的常数倍。

2. 对固定邻边使用面积界，或用容纳关系比较高。

3. 明确等号角，并证明原可行构型能达到它。

> Example. Points P, Q lie on concentric circles of radii two and three. Find the greatest area of triangle OPQ.

> 例题：P, Q 分别位于圆心为 O、半径二与三的同心圆上，求 OPQ 最大面积。

Worked solution

1. Area is $\frac{1}{2}(2)(3)|\sin\angle POQ| = 3|\sin\angle POQ|$.

2. This is at most three.

3. Choose perpendicular radii OP, OQ; both points remain on their circles, so area three is attainable.

Section 2.2 Areas & area ratios

面积与面积比

Method 10 / 方法 10 [GE02-M10] (continued / 续)

例题解析

1. 面积为 $3|\sin\angle POQ|$。

2. 因此面积至多三。

3. 取两半径垂直，两点仍位于各自圆上，故面积三确能达到。

An equality angle may be excluded by a segment or non-overlap condition.

等号角可能被线段或不重叠条件排除。

Practice / 配套练习: AIME-2006-I-Q08 (WB p. 250), AIME-2011-I-Q08 (WB p. 251), AIME-2013-II-Q10 (WB p. 252)

Section 2.2 Areas & area ratios

面积与面积比

Method 11 / 方法 11 [GE02-M11]

Central symmetry pairs equal area on opposite sides of a line

中心对称把直线两侧面积配对

Procedure

1. Identify pieces exchanged by a half-turn about a point on the proposed bisector.

2. The half-turn maps one open side of the line to the other and preserves area.

3. Handle remaining uncut pieces by equal-area pairing, then add the disjoint contributions.

4. For overlap polygons, intersect every boundary half-plane before measuring the result.

通用步骤

1. 找出关于拟议平分线上一点半周旋转后互换的部分。

2. 半周旋转交换直线两开半平面并保持面积。

3. 其余未切块按等面积配对，再相加互不重叠的贡献。

4. 重叠多边形须先相交全部边界半平面，再测量所得区域。

> Example. Two unit disks are centered at $(-1,0),(1,0)$. Show every line through the origin bisects the area of their union.

> 例题：两个单位圆盘圆心为 $(-1,0),(1,0)$，证明任何过原点直线都平分它们并集的面积。

Worked solution

1. A half-turn about the origin exchanges the two disks and preserves their union.

Section 2.2 Areas & area ratios

面积与面积比

Method 11 / 方法 11 [GE02-M11] (continued / 续)

2. For any line through the origin, the same half-turn exchanges its two sides and preserves area.

3. The line itself has area zero, so each side receives half the union area, namely π.

例题解析

1. 关于原点半周旋转交换两圆盘，并保持并集。

2. 对任一过原点直线，同一旋转交换两侧且保持面积。

3. 直线本身面积零，所以两侧各占并集一半，即 π。

A line can bisect a pair’s total area without bisecting each component.

一条线可以平分两个部件的总面积，而不分别平分每个部件。

Practice / 配套练习: AIME-2006-I-Q10 (WB p. 369), AIME-2021-I-Q02 (WB p. 361)